

Xijing Wang

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Education

Carnegie Mellon University, School of Computer Science

M.S. in Automated Science | Director's Fellowship

Pittsburgh, PA

Aug 2025 – May 2027

Santa Clara University, College of Arts and Sciences

B.S. in Computer Science (Algorithms), Minor in Economics | REAL Program Scholar

Santa Clara, CA

Sep 2021 – Mar 2025

Technical Skills

Languages: Python, C++, Swift, Java, Go, Scala

ML / MLSys: PyTorch, TensorFlow, Core ML, coremltools, MLX, TVM/MLC-LLM, Transformers, Faiss

Infrastructure: End-to-End ML Training, Model Conversion, On-device LLM Inference, Model Quantization, **LLM Serving**, RAG, HPC, Linux/Shell

Experience

Apple

Cupertino, CA

Software Engineer Intern | Python, Core ML, coremltools, MLX, Swift

May 11 – Aug 21, 2026

- Built an **agentic end-to-end ML training workflow** for cellular ML models, connecting training, evaluation, export, and packaging into **Core ML package** artifacts deployable to iPhone for device-side performance testing.
- Built **next-stage extensions of Call Context**, the feature announced at **WWDC26**, using **on-device LLM inference**; managed session/prewarm, runtime invocation, cross-functional internal infrastructure integration, and iPhone-side validation/performance.
- Upstreamed Apple ML ecosystem contributions: merged a **LightGBM converter** for **coremltools** [PR #2731] and **MLX** export/tracing support for custom Metal kernels from non-Metal builds [PR #3650].

Open Source Contributor — MLC-LLM & Apache TVM

Remote

Embedding model serving and compiler/runtime support | Python, C++, TVM

Feb 2026 – Present

- Merged OpenAI-compatible **/v1/embeddings** serving in **MLC-LLM** [PR #3430], adding Qwen3/BGE pooling, normalization, endpoint tests, and cross-backend cosine >0.999 vs. HF/Ollama.
- Merged **greedy sub-batching** for decoder embeddings [PR #3436], reducing p50 latency by **14×** (36s→2.5s); upstreamed **Apache TVM** masked sequence prefill support [PR #19392][#19431].

Research Engineer (AI Agent)

Pittsburgh, PA

Carnegie Mellon University | Python, FastAPI, Transformers

Jul 2025 – Dec 2025

- Built a **RAG**-based biomedical literature analysis platform with parallel retrieval from **PubMed** and **MedRxiv**, supporting Quick (<30s) and Deep (<3min) structured analysis modes.
- Fine-tuned **Qwen3 0.6B embedding model** on **8× A100 80GB** GPUs for biomedical domain adaptation; poster at **NeurIPS 2025 GenAI4Health Workshop**.

Machine Learning Research Intern

Santa Clara, CA

Lab of Cognitive & Computational Neuroscience, Santa Clara University | Python, TensorFlow, Shell

Jun 2024 – Sep 2024

- Developed **CNN** models and post-trained on **VGGFace2** (10K images) to validate generalizability of E/I imbalance theory in autism spectrum disorder (ASD) research.
- Built **fMRI** preprocessing on **HPC** and Pearson-correlation analysis workflows; first-author paper accepted at **CogSci 2025** and published in *Communications Biology*.

Projects

LocalMelo — Localized AI Agent Framework | *Python, FastAPI, LLM Serving* [GitHub]

2026

- Designed **LocalMelo**, a **local-first** agent framework that separates LLM serving support from the agent runtime, with **planner**, **memory**, **executor**, and **sleep** components for localized AI workflows.

Headache Note — On-device LLM Health Assistant | *MLC-LLM, Swift, SwiftUI, PyTorch* [GitHub]

2025

- Built an offline iOS AI agent by quantizing **LLaMA 3** (1.6B) with **MLC-LLM** for on-device inference on iPhone; used **SwiftData** for private, personalized health recommendations.

OpenOT2 — Lab Automation Framework | *Python, OpenCV, YOLO, LLM, REST API* [GitHub]

2025

- Open-sourced a modular **Python** framework for OT-2 liquid handling robots; built a stateful **HTTP** client with session management and exponential backoff retries for reliable robot communication over **REST API**.
- Integrated real-time **YOLO**-based computer vision pipeline with **OpenCV** USB camera capture for automated tip/liquid verification; added **LLM**-powered natural-language protocol generation with dry-run simulation; adopted by research labs at CMU.

Publications

[NeurIPS '25] Poster at **GenAI4Health Workshop**, San Diego, CA, Dec 2025

[CogSci '25] Poster at 47th Annual Meeting of the Cognitive Science Society, San Francisco, CA, Jul 2025

[Commun Biol '26] Xijing Wang, Emily Rios, and Lang Chen. “E/I imbalance and internal noise cause weak neural representations and face recognition challenges in ASD.” *Communications Biology*. [DOI]